

A SIMULATOR FOR OIL AND REFINED PRODUCTS SPOT PRICES USING A MULTIVARIATE STATE SPACE MODEL

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Abstract

In this paper we develop a multivariate state space model for oil and refined products spot prices based on a two-factor financial model originally developed in Schwartz and Smith [18]. Using the fundamentals of this two-factor model and the multivariate framework of Diebold et al. [5], we develop our multivariate model using the following strategy: for each refined product, other than natural gas, the log of spot price has its associated short and long term factor, being each of them an affine function of crude oil short term and long term factor, respectively. Therefore, in our model, the two crude oil factors function are the “global factors” driving the refined products spot prices. Model estimation is accomplished by use of the Kalman filter and maximum likelihood, with exact diffuse initialization, missing values treatment and univariate treatment for multivariate series.

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1. Introduction

The price of oil along with interest rates and exchange rates are some of the risk factors affecting decision-making processes in the oil industry. For example, when evaluating the feasibility of oil exploration and production projects it is particularly relevant to obtain simulated future price paths for oil and some of its derived products so that future cash flows originated from these investments can be estimated (Geman, 2005, cap. 9; Schwartz and Smith [18]). In the context of commodity pricing, models are usually specified as continuous time stochastic equations in which the observed time series is driven by a set of unobserved components or factors, which have financial and/or economic justifications (Schwartz and Smith [18]). The specification is completed by assuming particular stochastic processes for the evolution of the factors over time and discrete time equations are considered only at the estimation stage; the latter can be implemented under a *state space modeling* framework. Hence its accompanying estimation methods – the *Kalman filtering* and derived algorithms such as smoothing and maximum likelihood by the prediction error decomposition – can be used for factors extraction, fixed parameters estimation, prediction and simulation.

In the original two-factor model of Schwartz and Smith [18] (henceforth SS model), the log of oil spot price is decomposed in a long run and a short run factor. The long run factor represents the equilibrium price, which by assumption is driven by a geometric Brownian motion with a deterministic drift. This component should capture persistent changes in oil prices, like expectations of oil exhaustion, inflation etc. By its turn, the short term factor is modeled by a mean reversion process (Ornstein-Uhlenbeck process) reverting to zero. These deviations around the long-run trend are mainly due to transient features on the demand and supply sides, such as a severe winter in the northern hemisphere.

In our extension of SS model, we assume that the drift of the long run factor is no longer fixed in time but now follows a mean reversion process. This results in a forecasting function for the log of spot price that gradually approaches an upper bound, which looks sounder than predictions with unbounded growth. Secondly, thanks to co-integration among oil and most of refined product prices series (petrol, diesel, heating oil, aviation fuel etc), it is

possible to construct a multivariate extension of the SS model where only two common factors are used to drive the movements of each of the products price series, other than natural gas. Such framework presents some natural features desirable for our purposes of prediction and simulation, namely: it is based solely on past price data and some financial reasoning; it incorporates, whenever available, prices from futures contracts at different maturities; it deals easily with missing values occurring, mainly in prices series at long maturities; it allows one to deal with multivariate oil price models under a univariate treatment, something that is extremely expedient in terms of computational efficiency; and it is able to forecast and simulate future and spot prices for oil and its refined products.

The rest of the paper is organized as follows. In Section 2, for the sake of completeness, we review the main results of the SS model, giving emphasis on its corresponding forecasting equations. In Section 3, still within the univariate framework of the two factors model, we add a third factor to the original the SS model, by considering a stochastic drift on the long run factor. As before, we focus on the forecasting equations for both the spot and future prices associated with this new model. Finally, in Section 4 we develop our multivariate model for the joint forecast of spot prices for oil and the following products: gasoline, jet fuel, heating oil, naphta, diesel, propane, butane, and natural gas. Section 5 presents the results of model fitting to real data. Conclusions are given in Section 6.

2. SS Univariate Two-Factor Model

Formally, the univariate SS model is given by two set of equations. The first equation relates the log of oil spot prices to two factors: the short term factor χ_t , modeled by a mean reversion process (Ornstein-Uhlenbeck process) reverting to zero, and the long run factor ξ_t , driven by a geometric Brownian motion with a drift

$$\ln S_t = \chi_t + \xi_t. \quad (1)$$

The equations that define the evolution of the short term (χ_t) and the long term factors (ξ_t), known as the state equation, are given by the following continuous time stochastic processes:

$$\begin{aligned}
d\chi_t &= -\kappa\chi_t + \sigma_\chi dz_\chi \\
d\xi_t &= \mu_\xi dt + \sigma_\xi dz_\xi,
\end{aligned} \tag{2}$$

with $dz_\chi dz_\xi = \sigma_\chi \sigma_\xi \rho_{\chi\xi} dt$; meaning of each of the parameters is given in Table 1.

Table 1. SS model parameters description.

parameter	description
σ_χ	short-term volatility
σ_ξ	long-term volatility
μ_ξ	equilibrium drift rate
κ	short-term mean-reversion rate
$\rho_{\chi\xi}$	correlation in increments

It should be noticed that the two factors in the SS model are non-orthogonal since the shocks that drive them are correlated, by assumption. The SS model as such is not estimable, since the spot price and both components are unobservable. Observable oil prices are nevertheless available in futures markets (NYMEX, for example), at different maturities τ (e.g., 1 month, 24 months, 36 months etc). Following Duffie [3] and Schwartz and Smith [18], an appropriate mechanism to find out about the evolution of the unobserved oil spot price is to assume that, for any particular period t , the conditional expected value of the spot price at horizon $\tau + t$ with respect to the information available until time t is given by the futures price at that horizon; the conditional expectation is taken with respect to the risk neutral measure. Formally $F_{t,\tau} = \mathbf{E}^*(S_{t+\tau} | \mathfrak{J}_t)$, where $\mathfrak{J}_t$ is the σ -field mentioned above. This will make possible to express the spot price model in terms of the future observable prices, resulting in an estimable model. In the sequel we derive such a model.

2.1. SS model in terms of futures prices

The SS model specified in terms of futures prices is derived by using the expression for the expected value of the spot price taken with respect to a risk neutral measure. Assuming that the factors are normally distributed, it follows that S_t will have conditional log normal distributions with expected values directly obtained in terms of the conditional first- and second- order moments of $\ln S_t$. Formally

$$F_{t,\tau} = \mathbf{E}^*(S_{t+\tau} | \mathfrak{J}_t) = \exp(\mathbf{E}^*(\ln S_{t+\tau} | \mathfrak{J}_t) + \frac{1}{2} \mathbf{Var}^*(\ln S_{t+\tau} | \mathfrak{J}_t)), \quad (3)$$

where $\mathbf{E}^*$ e $\mathbf{Var}^*$ are the mean and variance of $\ln S_t$ obtained from the factor equations in its risk neutral form. For the moment we will leave these equations as they stand and will concentrate on obtaining the risk neutral form for the factor equations. This, as shown in Schwartz and Smith [18], is obtained by just adding a risk premium parameter to each equation, resulting in:

$$\begin{aligned} d\chi_t &= (-\kappa\chi_t - \lambda_\chi)dt + \sigma_\chi dz_\chi^*, \\ d\xi_t &= (\mu_\xi - \lambda_\xi)dt + \sigma_\xi dz_\xi^*, \end{aligned} \quad (4)$$

with $dz_\chi^* dz_\xi^* = \sigma_\chi \sigma_\xi \rho_{\chi\xi} dt$. Using Euler discretization method (T.-Y. Li and J. A. Yorke [22]; M. Yamaguti and H. Matano [14]; M. Hata [13]), the discrete time form for the short and long term factors equations are given by:

$$\begin{aligned} \chi_t &= -\lambda_\chi \Delta t + e^{-\kappa\Delta t} \chi_{t-\Delta t} + \omega_{1,t}^*, \\ \xi_t &= \mu_\xi^* \Delta t + \xi_{t-\Delta t} + \omega_{2,t}^*, \end{aligned} \quad (5)$$

with $\mu_\xi^* = (\mu_\xi - \lambda_\xi)$. The parameters λ_χ and λ_ξ are the risk premium for the short and long run factors, respectively; also, $\omega_{1,t}^* = \sigma_\chi \Delta z_\chi^*$, $\omega_{2,t}^* = \sigma_\xi \Delta z_\xi^*$ and $\mathbf{Corr}(\omega_{1,t}^*, \omega_{2,t}^*) = \sigma_\chi \sigma_\xi \rho_{\chi\xi} \Delta t$.

Finally, the equation for the log of futures prices at maturity τ , at time t , is given by

$$\ln F_{t,\tau} = A^*(\tau) + e^{-\kappa\tau} \chi_t + \xi_t + v_{t,\tau}, \quad v_{t,\tau} \sim N(0, \sigma_{v_{t,\tau}}^2), \quad (6)$$

with

$$A^*(\tau) = \mu_\xi^* \tau - \frac{\lambda_\chi}{\kappa} (1 - e^{-\kappa\tau}) + \frac{1}{2} \left((1 - e^{-2\kappa\tau}) \frac{\sigma_\chi^2}{2\kappa} + \sigma_\xi^2 \tau + 2 \frac{\sigma_\chi \xi_\xi \rho_{\chi\xi}}{\kappa} (1 - e^{-\kappa\tau}) \right),$$

with the real equations factor given in discrete time:

$$\begin{aligned} \chi_t &= e^{-\kappa\Delta t} \chi_{t-\Delta t} + \omega_{1,t} \\ \xi_t &= \mu_\xi \Delta t + \xi_{t-\Delta t} + \omega_{2,t}, \end{aligned} \quad (7)$$

where $\omega_{1,t} = \sigma_\chi \Delta z_\chi$, $\omega_{2,t} = \sigma_\xi \Delta z_\xi$ and $\mathbf{Corr}(\omega_{1,t}, \omega_{2,t}) = \sigma_\chi \sigma_\xi \rho_{\chi\xi} \Delta t$.

2.2. Forecasting of the spot price and futures prices

In this section we obtain the 1 steps ahead forecasting function for both the unobserved spot price $S(t)$ and for the future price $F(t)$ at any maturity. Although primarily unobservable, current and future values for the spot prices can be estimated from the SS model in futures prices form. To simulating oil prices, the spot price should be used instead of any of the future prices. This choice is justified by the fact that the estimated spot price can be seen as a weighted average of the futures prices at different maturities, and thus a carrier of future market price information. In the sequel, we consider the forecasting for $t = T + \ell\Delta t$, with $\ell = 1, 2, \dots$, and T is the last observation available for the series of futures prices. The expression for the projected conditional mean of the *spot price*, considering $\Delta t \approx 0$, is given by:

$$\begin{aligned} \hat{S}_{T+\ell} &= \mathbf{E}(S_{T+\ell} | J_T) \\ \hat{S}_{T+\ell} &= \exp \left(\mathbf{E}(S_{T+\ell} | J_T) + \frac{1}{2} \mathbf{Var}(S_{T+\ell} | J_T) \right) \\ \hat{S}_{T+\ell} &= \exp(\hat{A}(\ell) + e^{-\hat{\kappa}\ell\Delta t} \hat{\chi}_T + \hat{\xi}_T) \end{aligned} \quad (8)$$

with

$$\hat{A}(\ell) = \hat{\mu}_\xi \ell\Delta t + \frac{1}{2} \left(\frac{1 - e^{-2\hat{\kappa}\ell\Delta t}}{2\hat{\kappa}} \hat{\sigma}_\chi^2 + \hat{\sigma}_\xi^2 \ell\Delta t + 2 \frac{1 - e^{-\hat{\kappa}\ell\Delta t}}{\hat{\kappa}} \hat{\rho}_{\chi\xi} \hat{\sigma}_\chi \hat{\sigma}_\xi \right).$$

The asymptotic behavior of this expression can be obtained by making ℓ large, and considering $0 < (1 - \hat{\kappa}\Delta t) < 1$. Under these conditions the forecasting function for the spot price becomes:

$$\begin{aligned}\hat{S}_{T+\ell} &= \exp(\hat{A}(\ell)_\infty + \hat{\xi}_T) \\ \hat{A}(\ell)_\infty &= \left(\hat{\mu}_\xi \frac{\hat{\sigma}_\xi^2}{2} \right) \ell \Delta t + \frac{1}{2} \left(\frac{\hat{\sigma}_\chi^2}{2\hat{\kappa}} + 2 \frac{\hat{\rho}_{\chi\xi} \hat{\sigma}_\chi \hat{\sigma}_\xi}{\hat{\kappa}} \right).\end{aligned}\quad (9)$$

From equation (9) one can easily see that the asymptotic forecasting function for the spot price is the exponential of a linear trend, where the slope is determined solely by parameters associated with the long run factor: the long-term volatility $\hat{\mu}_\xi$ and the equilibrium drift rate $\hat{\sigma}_\xi^2$.

In a similar fashion one can show that the ℓ steps ahead forecasting function for the future price at any maturity τ is given by

$$\begin{aligned}\hat{F}_{T+\ell}^\tau &= \mathbf{E}^*(F_{T+\ell} | J_T) \\ \hat{F}_{T+\ell}^\tau &= \exp(\mathbf{E}^*(\ln F_{T+\ell} | J_T) + \frac{1}{2} \mathbf{Var}^*(\ln F_{T+\ell} | J_T)) \\ \hat{F}_{T+\ell}^\tau &= \exp(\hat{A}^*(\tau + \ell \Delta \tau) + e^{-\hat{\kappa}(\tau + \ell \Delta t)} \hat{\chi}_T + \hat{\xi}_T + \frac{\hat{\sigma}_{v_\tau}}{2})\end{aligned}\quad (10)$$

with

$$\begin{aligned}\hat{A}^*(\tau + \ell \Delta t) &= -\frac{\hat{\lambda}_\chi}{\hat{\kappa}} (1 - e^{-\hat{\kappa}(\tau + \ell \Delta t)}) + \mu_\xi^*(\tau + \ell \Delta t) + \dots \\ &+ \frac{1}{2} \left(\frac{1 - e^{-2\hat{\kappa}(\tau + \ell \Delta t)}}{2\hat{\kappa}} \hat{\sigma}_\chi^2 + \hat{\sigma}_\xi^2 (\tau + \ell \Delta t) + 2 \frac{1 - e^{-\hat{\kappa}(\tau + \ell \Delta t)}}{\hat{\kappa}} \hat{\rho}_{\chi\xi} \hat{\sigma}_\chi \hat{\sigma}_\xi \right).\end{aligned}$$

For ℓ large, this expression becomes:

$$\hat{F}_{T+\ell}^\tau = \exp \left(\hat{A}_\infty^*(\tau + \ell \Delta t) + \hat{\xi}_T + \frac{\hat{\sigma}_{v_\tau}}{2} \right), \quad (11)$$

where

$$\hat{A}_\infty^*(\tau + \ell \Delta t) = -\frac{\hat{\lambda}_\chi}{\hat{\kappa}} + \left(\hat{\mu}_\xi^* + \frac{\hat{\sigma}_\xi^2}{2} \right) (\tau + \ell \Delta t) + \frac{1}{2} \left(\frac{\hat{\sigma}_\chi^2}{2\hat{\kappa}} + 2 \frac{\hat{\rho}_{\chi\xi} \hat{\sigma}_\chi \hat{\sigma}_\xi}{\hat{\kappa}} \right).$$

In both forecasting functions, we have used that Δt is small, which make it possible to consider the approximation $(1 - \kappa\Delta t) \approx e^{-\kappa\Delta t}$.

2.3. Forecasting spot prices through simulation

We now show how to use the SS model to obtain simulated spot prices for any future horizon $T + \ell$, $\ell > 0$. These will be key inputs for procedures developed to asses future cash flows derived from oil exploration and production projects, as for example, in corporate asset liability management (Geman, 2005, cap. 9). In order to implement this simulation procedure, one only needs the estimated factors at the last period for which observations are available and random numbers generated from the multivariate normal distribution associated with the noise vector that is part of the stochastic equations in (7). This simulation scheme will also be valid when considering the proposed multivariate model. The details of the simulation are shown in Figure 1.

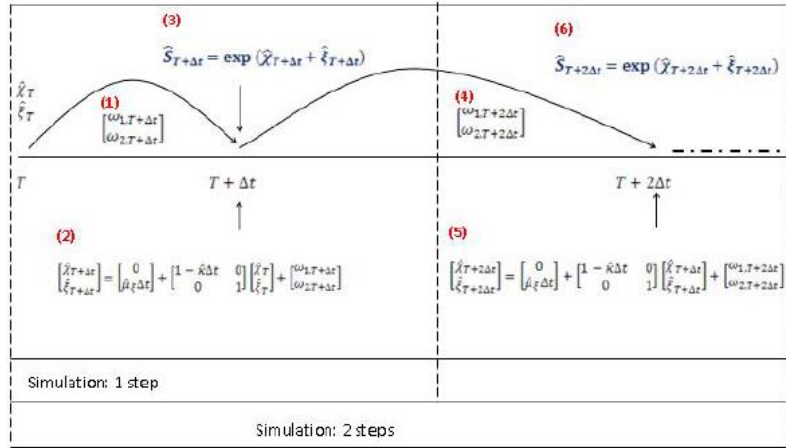


Figure 1. Price simulation using the SS model.

3. SS Model with a Stochastic Drift

As shown in equations (9) and (10), the forecasting function for both spot and futures prices may diverge with ℓ , the time horizon. This could result in highly non-stationary simulated paths for oil and refined product prices when considering very long time horizons. One way to partially reconcile price

mean reversion with the Brownian motion process that leads the forecasting function in the SS model is to allow the Brownian motion growth rate (μ_ξ) to vary stochastically in time, following a mean reversion of its own. As a result the trend for the expected log prices will converge gradually to a fixed value. This modification has been considered in Schwartz and Smith [17], and incidentally has a counterpart in the time series literature, known as the damped (stochastic) trend model, as presented in the context of unobserved components models by Harvey [7], ch. 2, p. 46. In the sequel we present the equations for SS stochastic drift model where the drift of the long run process follows a mean reversion equation, resulting in a damped trend model for the log of spot price.

$$\begin{aligned}
\ln S_t &= \chi_t + \xi_t \\
d\chi_t &= -\kappa_\chi \chi_t dt + \sigma_\chi dz_\chi \\
d\xi_t &= \mu_t dt + \sigma_\xi dz_\xi \\
d\mu_t &= -\kappa_\mu (\mu_t - \bar{\mu}) dt + \sigma_\mu dz_\mu,
\end{aligned} \tag{12}$$

with $dz_\chi dz_\xi = \rho_{\chi\xi} dt$, $dz_\chi dz_\mu = \rho_{\chi\mu} dt$ e $dz_\mu dz_\xi = \rho_{\xi\mu} dt$. By its turn, the risk neutral version of the previous equations (12) are given by:

$$\begin{aligned}
\ln S_t &= \chi_t + \xi_t \\
d\chi_t &= (-\kappa_\chi \chi_t - \lambda_\chi) dt + \sigma_\chi dz_\chi^* \\
d\xi_t &= (\mu_t - \lambda_\xi) dt + \sigma_\xi dz_\xi^* \\
d\mu_t &= (-\kappa_\mu (\mu_t - \bar{\mu}) + \lambda_\mu) dt + \sigma_\mu dz_\mu^*
\end{aligned} \tag{13}$$

where λ_χ , λ_ξ and λ_μ are the short-term factor, long-term factor and drift risk premium, respectively; $dz_\chi^* dz_\xi^* = \rho_{\chi\xi} dt$, $dz_\chi^* dz_\mu^* = \rho_{\chi\mu} dt$ e $dz_\mu^* dz_\xi^* = \rho_{\xi\mu} dt$.

The discrete time version of the previous equations (12) and (13) are given by:

$$\chi_t = (1 - \kappa_\chi \Delta t) \chi_{t-\Delta t} + \sigma_\chi \Delta z_\chi$$

$$\xi_t = \mu_{t-\Delta t} \Delta t + \xi_{t-\Delta t} + \sigma_\xi \Delta z_\xi$$

$$\mu_t = (1 - \kappa_\mu \Delta t) \mu_{t-\Delta t} + \kappa_\mu \bar{\mu} \Delta t + \sigma_\mu \Delta z_\mu \quad (14)$$

$$\chi_t = -\lambda_\chi \Delta t + (1 - \kappa_\chi \Delta t) \chi_{t-\Delta t} + \sigma_\chi \Delta z_\chi^*$$

$$\xi_t = -\lambda_\xi \Delta t + \mu_{t-\Delta t} \Delta t + \xi_{t-\Delta t} + \sigma_\xi \Delta z_\xi^*$$

$$\mu_t = (1 - \kappa_\mu \Delta t) \mu_{t-\Delta t} + \kappa_\mu \bar{\mu}^* \Delta t + \sigma_\mu \Delta z_\mu^*, \quad (15)$$

where $\bar{\mu}^* = \left(\bar{\mu} - \frac{\lambda_\mu}{\kappa_\mu} \right)$, $\Delta z_\chi \Delta z_\xi = \Delta z_\chi^* \Delta z_\xi^* = \rho_{\chi\xi} \Delta t$, $\Delta z_\chi \Delta z_\mu = \Delta z_\chi^* \Delta z_\mu^* = \rho_{\chi\mu} \Delta t$

and $\Delta z_\mu \Delta z_\xi = \Delta z_\mu^* \Delta z_\xi^* = \rho_{\xi\mu} \Delta t$.

3.1. SS stochastic drift model using future prices

As in the original SS model, we have to obtain the stochastic drift model in terms of futures prices. From equations (15) and the relation (3), one can arrive at the following equation for the future prices at maturity τ in time t :

$$\ln F_{t,\tau} = B^*(\tau) + e^{-\kappa_\chi \tau} \chi_t + \xi_t + \frac{1 - e^{-\kappa_\mu \tau}}{\kappa_\mu} \mu_t + v_{t,\tau}, \quad (16)$$

where $v_{t,\tau} \sim N(0, \sigma_{v_{t,\tau}})$ and

$$B^*(\tau) = -\frac{\lambda_\chi}{\kappa_\chi} (1 - e^{-\kappa_\chi \tau}) - \lambda_\xi \tau + \bar{\mu}^* \left(\tau - \frac{1 - e^{-\kappa_\mu \tau}}{\kappa_\mu} \right) + \frac{1}{2} (s_{11}(\tau) + 2s_{12}(\tau) + s_{22}(\tau)),$$

where

$$s_{11}(\tau) = (1 - e^{-2\kappa_\chi \tau}) \frac{\sigma_\chi^2}{2\kappa_\chi}$$

$$s_{22}(\tau) = \sigma_\xi^2 \tau + \frac{\rho_{\xi\mu} \sigma_\xi \sigma_\mu}{\kappa_\mu} \left(\tau - \frac{1 - e^{-2\kappa_\mu \tau}}{\kappa_\mu} \right) + \frac{\sigma_\mu^2}{\kappa_\mu^2} \left(\tau - \frac{2}{\kappa_\mu} (1 - e^{-\kappa_\mu \tau}) + \frac{1}{2\kappa_\mu} (1 - e^{-2\kappa_\mu \tau}) \right)$$

$$s_{12}(\tau) = \rho_{\chi\xi} \sigma_\chi \sigma_\xi \left(\frac{1 - e^{-\kappa_\chi \tau}}{\kappa_\chi} \right) + \frac{\rho_{\chi\mu} \sigma_\chi \sigma_\mu}{\kappa_\mu} \left(\frac{1 - e^{-\kappa_\mu \tau}}{\kappa_\chi} - \frac{1 - e^{-(\kappa_\chi + \kappa_\mu) \tau}}{\kappa_\chi + \kappa_\mu} \right)$$

3.2. Forecasting of the spot price

The forecasting function ℓ steps ahead for this new model can also be obtained using the first two equations in (8), resulting in:

$$\hat{S}_{T+\ell} = \exp \left(\hat{B}(\ell) + e^{-\hat{\kappa}_\chi \ell \Delta t} \hat{\chi}_T + \hat{\xi}_T + \frac{1 - e^{-\hat{\kappa}_\mu \ell \Delta t}}{\hat{\kappa}_\mu} \hat{\mu}_T \right), \quad (17)$$

where

$$\hat{B}(\ell) = \bar{\mu} \left(\ell \Delta t - \frac{1 - e^{-\kappa_\mu \ell \Delta t}}{\kappa_\mu} \right) + \frac{1}{2} (s_{11}(\ell \Delta t) + 2s_{12}(\ell \Delta t) + s_{22}(\ell \Delta t)).$$

When ℓ is large, the previous expression can be written as:

$$\hat{S}_{T+\ell} = \exp \left(\hat{B}(\ell)_\infty + \hat{\xi}_T + \frac{\hat{\mu}_T}{\hat{\kappa}_\mu} \right), \quad (18)$$

where

$$\begin{aligned} \hat{B}(\ell)_\infty = & -\frac{\bar{\mu}}{\kappa_\mu} + \frac{1}{2} \left(\frac{\rho_{\xi\mu} \sigma_\xi \sigma_\mu}{\kappa_\mu^2} + \frac{\hat{\sigma}_\chi^2}{2\hat{\kappa}_\chi} + 2 \frac{\hat{\rho}_{\chi\xi} \hat{\sigma}_\chi \hat{\sigma}_\xi}{\hat{\kappa}_\chi} - \frac{3\hat{\sigma}_\chi^2}{2\hat{\kappa}_\mu^3} + 2 \frac{\rho_{\chi\mu} \sigma_\chi \sigma_\mu}{\kappa_\chi (\kappa_\chi + \kappa_\mu)} \right) \\ & + \left(\frac{\hat{\sigma}_\xi^2}{2} + \frac{\hat{\sigma}_\mu^2}{2\hat{\kappa}_\mu^2} + \bar{\mu} + \frac{\rho_{\xi\mu} \sigma_\xi \sigma_\mu}{\kappa_\mu} \right) \ell \Delta t. \end{aligned}$$

It should be noticed that the stochastic drift model produces a long term fixed mean for the log of spot prices. In fact when $\ell \rightarrow \infty$ the forecasting function for $\ln S_t$ converges to $\hat{\xi}_T + \frac{\hat{\mu}_T}{\hat{\kappa}_\mu}$. Nevertheless, under the damped trend model the forecasting function for the spot price may still diverge when $\ell \rightarrow \infty$, since the trend slope $\left(\frac{\hat{\sigma}_\xi^2}{2} + \frac{\hat{\sigma}_\mu^2}{2\hat{\kappa}_\mu^2} + \bar{\mu} + \frac{\rho_{\xi\mu} \sigma_\xi \sigma_\mu}{\kappa_\mu} \right)$ can be positive or negative.

4. A Multivariate Model for Oil and Refined Products Spot Prices

4.1. Introduction

In the previous sections, we have presented the the SS three-factor model for the univariate modeling of oil spot price series. Here we develop a

multivariate extension of the SS stochastic drift model using the framework of a common factor model to be presented shortly. Instead of considering a full multivariate model in which each commodity series has its own three factors, we choose to adopt a more parsimonious specification. Specifically, the model is made up of only three common components, the very same used in the SS univariate model, namely, a short term factor, a long term factor, and the drift's long term factor. In the sequel we offer some empirical/theoretical justifications for the choice of such parsimonious model. Figure 2 presents the plot of price series for oil (WTI), gasoline and jet fuel in the period from January 1985 to September 2009. A simple visual inspection of these series suggests a non-stationary behavior for all three price series, but somehow their movements in time seem to evolve in synchrony, which is not a surprise, given that crude oil is the raw material for these products. It is then reasonable to conclude that part of the variation of the products series can be explained by the specific factors that drive the oil price series. In other words, these series share common factors. The presence of these common factors is what justify a parsimonious specification for the multivariate model of oil and its products series. Under these circumstances it can also be shown that these series will co-integrate, that is, there will be a number of linear combinations of the series which will be stationary, although the series themselves are non-stationary (cf. Harvey [7], ch. 8). Empirical evidences on the presence of common trends in crude oil and refined products have also been reported by Mirantes et al. [12], while Lanza et al. [11] have shown that these series co-integrate. A particular multivariate framework suited to our common factors strategy for the joint modeling of the price series of oil and its products is that of Diebold et al. [5] (henceforth DLY). DLY propose a state space model for the joint modeling of N -country yield curve by linking them to few global factors using affine relationships. We have adapted such framework to specify our multivariate model, as follows: for each refined product, other than natural gas, the log of spot price has its associated short and long term factor, being each of them an affine function of crude oil (WTI) short term and long term factor, respectively. Therefore, in our model, the two crude oil factors represent the “global factors” driving the refined products spot prices. Given the idiosyncracies of natural gas price, which is less prone to respond directly to variations in crude oil prices, a pair of factors was specifically set to its spot equation.

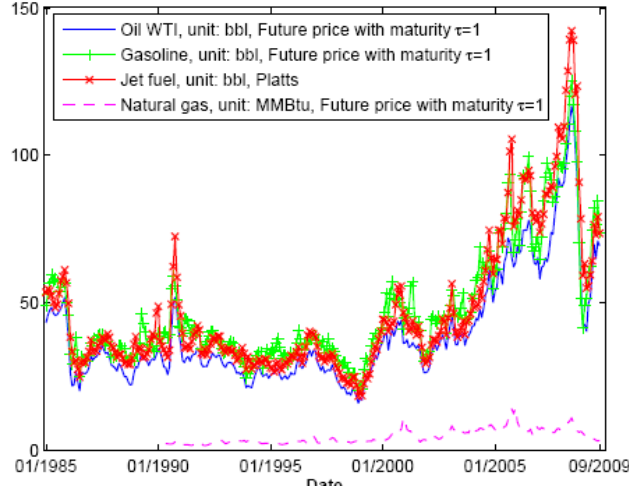


Figure 2. Future prices for oil and some of its products.

4.2. Model equations

Hereafter, for ease of notation the term product will refer to crude oil or one of its refined products. For each product i , we consider that the log of spot price follows the SS stochastic drift model, that is:

$$\begin{aligned}\ln S_{t,i} &= \chi_{t,i} + \xi_{t,i} \\ \chi_{t,i} &= \phi_{i,0}^X + \phi_{i,1}X_t + \epsilon_{t,i}^X \\ \xi_{t,i} &= \phi_{i,0}^E + \phi_{i,2}E_t + \epsilon_{t,i}^E,\end{aligned}\tag{19}$$

where X_t , E_t and U_t are the crude oil or global factors that drives the log of spot price for all refined products other than natural gas.

By substitution of the individual factors $\chi_{t,i}$ e $\xi_{t,i}$ in the equation for the (log) of spot prices, one gets:

$$\begin{aligned}\ln S_{t,i} &= \phi_{i,0} + \phi_{i,1}X_t + \phi_{i,2}E_t + \epsilon_{t,i} \\ dX_t &= -\kappa_X X_t dt + \sigma_X dz_X \\ dE_t &= U_t dt + \sigma_E dz_E \\ du_t &= -\kappa_U (U_t - \bar{U})dt + \sigma_U dz_U.\end{aligned}\tag{20}$$

We now add some restrictions to the parameters ϕ 's in order to guarantee that, for certain products, the simulated oil price will be smaller than the products price, and at the same time the spreads between oil and products will be a stationary process.

- oil ($i = 1$) : $\phi_{1,0} = 0$, $\phi_{1,1} = 1$ and $\phi_{1,2} = 1$
- gasoline ($i = 2$) : $\phi_{2,0} > 0$, $\phi_{2,1} > 1$ and $\phi_{2,2} = 1$
- jet fuel ($i = 3$) : $\phi_{3,0} > 0$, $\phi_{3,1} > 1$ and $\phi_{3,2} = 1$
- heating oil (LSFO) ($i = 4$) : $\phi_{4,0} < 0$, $\phi_{4,1} < 1$ and $\phi_{4,2} < 1$
- naphtha ($i = 5$) : $\phi_{5,0} > 0$, $\phi_{5,1} > 1$ and $\phi_{5,2} > 1$
- diesel ($i = 6$) : $\phi_{6,0} > 0$, $\phi_{6,1} > 1$ and $\phi_{6,2} = 1$
- propane ($i = 7$) : $\phi_{7,0} < 0$, $\phi_{7,1} < 1$ and $\phi_{7,2} = 1$
- butane ($i = 8$) : $\phi_{8,0} < 0$, $\phi_{8,1} < 1$ and $\phi_{8,2} = 1$
- natural gas ($i = 9$) : $\phi_{9,0} > 0$, $\phi_{9,1} > 1$ and $\phi_{9,2} > 1$

In our multivariate model, the price of natural gas is not considered to be directly affected by the factors that drive oil prices, and therefore, it will have its own factors. Nevertheless the shocks that drive oil and natural gas factors equations will be correlated. In a similar fashion the equations that describe the evolution of natural gas spot prices are given by:

$$\begin{aligned}
 \ln S_t^{gnp} &= X_t^{gnp} + E_t^{gnp} \\
 dX_t^{gnp} &= -\kappa_X^{gnp} X_t^{gnp} dt + \sigma_X^{gnp} dz_X^{gnp} \\
 dE_t^{gnp} &= U_t^{gnp} dt + \sigma_E^{gnp} dz_E^{gnp} \\
 dU_t^{gnp} &= -\kappa_U^{gnp} (U_t^{gnp} - \bar{U}^{gnp}) dt + \sigma_U^{gnp} dz_U^{gnp}
 \end{aligned} \tag{21}$$

where $\phi_{i,0} = \phi_{i,0}^X + \phi_{i,0}^E$ and $\epsilon_{t,i} = \epsilon_{t,i}^X + \epsilon_{t,i}^E$. The discrete version of equations (20) is given by:

$$\begin{aligned}
\ln S_{t,i} &= \phi_{i,0} + \phi_{i,1} X_t + \phi_{i,2} E_t + \epsilon_{t,i} \\
X_t &= e^{-\kappa_X \Delta t} X_{t-\Delta t} + \sigma_X \Delta z_X \\
E_t &= U_{t-\Delta t} \Delta t + E_{t-\Delta t} + \sigma_E \Delta z_E \\
U_t &= e^{-\kappa_U \Delta t} U_{t-\Delta t} + \kappa_U \bar{U} \Delta t + \sigma_U \Delta z_U
\end{aligned} \tag{22}$$

For natural gas the discrete version of equation (21) is given by:

$$\begin{aligned}
\ln S_t &= X_t^{gnp} + E_t^{gnp} \\
X_t^{gnp} &= e^{-\kappa_X^{gnp} \Delta t} X_{t-\Delta t}^{gnp} + \sigma_X^{gnp} \Delta z_X^{gnp} \\
E_t^{gnp} &= U_{t-\Delta t}^{gnp} \Delta t + E_{t-\Delta t}^{gnp} + \sigma_E^{gnp} \Delta z_E^{gnp} \\
U_t^{gnp} &= e^{-\kappa_U^{gnp} \Delta t} U_{t-\Delta t}^{gnp} + \kappa_U^{gnp} \bar{U}^{gnp} \Delta t + \sigma_U^{gnp} \Delta z_U^{gnp}
\end{aligned} \tag{23}$$

4.3. Multivariate SS stochastic drift model using future prices

To obtain the equations for the future price at maturity τ for the product i in time t , we first consider the risk neutral version of the discrete form of the log of spot price (20), which may be shown to take the following form:

$$\begin{aligned}
\ln S_{t,i} &= \phi_{i,0} + \phi_{i,1} X_t + \phi_{i,2} E_t + v_{t,i} \\
X_t &= -\lambda_X \Delta t + e^{-\kappa_X \Delta t} X_{t-\Delta t} + \sigma_X \Delta z_X^* \\
E_t &= -\lambda_E \Delta t + U_{t-\Delta t} \Delta t + E_{t-\Delta t} + \sigma_E \Delta z_E^* \\
U_t &= e^{-\kappa_U \Delta t} U_{t-\Delta t} + \kappa_U \bar{U}^* \Delta t + \sigma_U \Delta z_U^*
\end{aligned} \tag{24}$$

It then follows that the equation for the future price is given by:

$$\begin{aligned}
\ln F_{t,\tau}^{(i)} &= B_i^*(\tau) + \phi_{i,1} e^{-\kappa_X \tau} X_t + \phi_{i,2} E_t + \frac{\phi_{i,2}}{\kappa_U} (1 - e^{-\kappa_U \tau}) U_t + v_{t,\tau}^{(i)} \\
B_i^*(\tau) &= \phi_{i,0} - \frac{\lambda_X}{\kappa_X} \phi_{i,1} (1 - e^{-\kappa_X \tau}) + \phi_{i,2} (\bar{U}^* - \lambda_E) \tau - \phi_{i,2} \bar{U}^* \left(\frac{1 - e^{-\kappa_U \tau}}{\kappa_U} \right)
\end{aligned}$$

$$+ \frac{1}{2} (\phi_{i,1}^2 S_{11}(\tau) + 2\phi_{i,1}\phi_{i,2} S_{12}(\tau) + \phi_{i,2}^2 S_{22}(\tau)), \quad (25)$$

$$\text{where } \bar{U}^* = \left(\bar{U} - \frac{\lambda_U}{\kappa_U} \right)$$

$$S_{11}(\tau) = (1 - e^{-2\kappa_X \tau}) \frac{\sigma_X^2}{2\kappa_X}$$

$$S_{22}(\tau) = \sigma_E^2 \tau + \frac{\rho_{EU} \sigma_E \sigma_U}{\kappa_U} \left(\tau - \frac{1 - e^{-\kappa_U \tau}}{\kappa_U} \right) + \frac{\sigma_U^2}{\kappa_U} \left(\tau - \frac{2}{\kappa_U} (1 - e^{-2\kappa_U \tau}) + \frac{1}{2\kappa_U} (1 - e^{-2\kappa_U \tau}) \right)$$

$$S_{12}(\tau) = \rho_{XE} \sigma_X \sigma_E \left(\frac{1 - e^{-\kappa_X \tau}}{\kappa_X} \right) + \frac{\rho_{XU} \sigma_X \sigma_U}{\kappa_U} \left(\frac{1 - e^{-\kappa_X \tau}}{\kappa_X} - \frac{1 - e^{-(\kappa_X + \kappa_U) \tau}}{\kappa_X + \kappa_U} \right).$$

The corresponding risk neutral equations for natural gas model (21) is given by:

$$\ln S_t = X_t^{gnp} + E_t^{gnp}$$

$$X_t^{gnp} = -\lambda_X^{gnp} \Delta t + e^{-\kappa_X^{gnp} \Delta t} X_{t-\Delta t}^{gnp} + \sigma_X^{gnp} \Delta z_X^{gnp*}$$

$$E_t^{gnp} = -\lambda_E^{gnp} \Delta t + U_{t-\Delta t}^{gnp} \Delta t + E_{t-\Delta t}^{gnp} + \sigma_E^{gnp} \Delta z_E^{gnp*}$$

$$U_t^{gnp} = e^{-\kappa_U^{gnp} \Delta t} U_{t-\Delta t}^{gnp} + \kappa_U^{gnp} \bar{U}^{*gnp} \Delta t + \sigma_U^{gnp} \Delta z_U^{gnp*}, \quad (26)$$

while the equation for the natural gas futures price is given by:

$$\ln F_{t,\tau} = B^{gnp*}(\tau) + e^{-\kappa_X^{gnp} \tau} X_t^{gnp} + E_t + \frac{U_t^{gnp}}{\kappa_U^{gnp}} (1 - e^{-\kappa_U^{gnp} \tau}) + v_{t,\tau}$$

$$\begin{aligned} B^{gnp*}(\tau) = & -\frac{\lambda_X^{gnp}}{\kappa_X^{gnp}} (1 - e^{-\kappa_X^{gnp} \tau}) + (\bar{U}^{*gnp} - \lambda_E^{gnp}) \tau - \bar{U}^{*gnp} \left(\frac{1 - e^{-\kappa_U^{gnp} \tau}}{\kappa_U^{gnp}} \right) \\ & + \frac{1}{2} (s_{11}^{gnp}(\tau) + 2s_{12}^{gnp}(\tau) + s_{22}^{gnp}(\tau)), \end{aligned} \quad (27)$$

$$\text{where } \bar{U}^{*gnp} = \left(\bar{U}^{gnp} - \frac{\lambda_U^{gnp}}{\kappa_U^{gnp}} \right)$$

$$\begin{aligned}
s_{11}^{gnp}(\tau) &= (1 - e^{-2\kappa_X^{gnp}\tau}) \frac{(\sigma_X^{gnp})^2}{2\kappa_X^{gnp}} \\
s_{22}^{gnp}(\tau) &= (\sigma_E^{gnp})^2 + \frac{\rho_{EU}^{gnp} \sigma_E^{gnp} \sigma_U^{gnp}}{\kappa_U^{gnp}} \left(\tau - \frac{1 - e^{-\kappa_U^{gnp}\tau}}{\kappa_U^{gnp}} \right) + \frac{(\sigma_U^{gnp})^2}{(\kappa_U^{gnp})^2} \left(\tau - \frac{2}{\kappa_U^{gnp}} (1 - e^{-\kappa_U^{gnp}\tau}) \right. \\
&\quad \left. + \frac{1}{2\kappa_U^{gnp}} (1 - e^{-2\kappa_U^{gnp}\tau}) \right) \\
s_{12}^{gnp}(\tau) &= \sigma_{XE}^{gnp} \sigma_X^{gnp} \sigma_E^{gnp} \left(\frac{1 - e^{-\kappa_X^{gnp}\tau}}{\kappa_X^{gnp}} \right) + \frac{\rho_{XU}^{gnp} \sigma_X^{gnp} \sigma_U^{gnp}}{\kappa_U^{gnp}} \\
&\quad \times \left(\frac{1 - e^{-\kappa_X^{gnp}\tau}}{\kappa_X^{gnp}} - \frac{1 - e^{-(\kappa_X^{gnp} + \kappa_U^{gnp})\tau}}{\kappa_X^{gnp} + \kappa_U^{gnp}} \right).
\end{aligned}$$

4.4. Forecasting of the spot price for crude oil and refined products

The ℓ steps ahead forecasting function for the spot price of product i is given by:

$$\hat{S}_{T+\ell,i} = \exp(\hat{B}_i(\ell) + \hat{\phi}_{i,1} e^{-\hat{\kappa}_X \ell \Delta t} \hat{X}_T + \hat{\phi}_{i,2} \hat{E}_T + \frac{\hat{\phi}_{i,2}}{\hat{\kappa}_U} (1 - e^{-\hat{\kappa}_U \ell \Delta t}) \hat{U}_T), \quad (28)$$

where

$$\hat{B}_i(\ell) = \phi_{i,0} + \phi_{i,2} \bar{U} \left(\ell \Delta t - \frac{1 - e^{-\kappa_U \ell \Delta t}}{\kappa_U} \right) + \frac{1}{2} (S_{11}(\ell \Delta t) + 2S_{12}(\ell \Delta t) + S_{22}(\ell \Delta t)).$$

When ℓ is large, the previous expression can be written as:

$$\begin{aligned}
\hat{B}_i(\ell)_\infty &= \phi_{i,0} - \phi_{i,2} \frac{\bar{U}}{\kappa_U} + \frac{1}{2} \left(\phi_{i,2}^2 \left(\frac{\rho_{EU} \sigma_E \sigma_U}{\kappa_U^2} - \frac{3\hat{\sigma}_U^2}{2\hat{\kappa}_U^3} \right) + \phi_{i,1}^2 \frac{\hat{\sigma}_X^2}{2\hat{\kappa}_X} \right. \\
&\quad \left. + 2\phi_{i,1} \phi_{i,2} \left(\frac{\hat{\rho}_{XE} \hat{\sigma}_X \hat{\sigma}_E}{\hat{\kappa}_E} + \frac{\rho_{XU} \sigma_X \sigma_U}{\kappa_E (\kappa_X + \kappa_U)} \right) \right) \\
&\quad + \left(\phi_{i,2}^2 \left(\frac{\hat{\sigma}_E^2}{2} + \frac{\hat{\sigma}_U^2}{2\hat{\kappa}_U^2} + \frac{\rho_{EU} \sigma_E \sigma_U}{\kappa_E} \right) + \phi_{i,2} \bar{U} + \right) \ell \Delta t.
\end{aligned}$$

In a similar fashion one can also obtain a closed formula for the forecasting function for the future price at maturity τ for product i . This is given by:

$$\begin{aligned} \hat{F}_{T+\ell, \tau}^{(i)} = & \exp(\hat{B}_i^*(\tau + \ell\Delta t) + \hat{\phi}_{i,1} e^{-\hat{\kappa}_X(\tau+\ell\Delta t)} \hat{X}_T + \hat{\phi}_{i,2} \hat{E}_T \\ & + \frac{\hat{\phi}_{i,2}}{\hat{\kappa}_U} (1 - e^{-\hat{\kappa}_U(\tau+\ell\Delta t)}) \hat{U}_T) \end{aligned} \quad (29)$$

where $\hat{B}_i^*(\tau + \ell\Delta t)$ is defined as in equation (25).

4.5. SS Multivariate stochastic drift model in state space form

The *state space form* is a general framework that encompasses a variety of statistical time series models, like univariate and multivariate structural models, time-varying regression models, ARIMA models, vector auto-regressions etc. It is particularly suitable to address the estimation of special types of models where some of the unknown parameters evolve stochastically in time. The discrete form of the previous models presented in this paper – the SS model, the SS model with stochastic drift, and our multivariate extension of the SS model with stochastic drift, which are given along equation (6), (7), 14, (16), (22) and (25) – fit in this framework, since each of them has unobserved factors following specific stochastic processes.

We define a *Gaussian linear state space model* by the following measurement equation, state equation and initial state vector:

$$\begin{aligned} Y_t &= Z_t \alpha_t + d_t + \varepsilon_t, \quad \varepsilon_t \sim NID(0, H_t), \\ \alpha_{t+1} &= T_t \alpha_t + c_t + \eta_t, \quad \eta_t \sim NID(0, Q_t). \end{aligned} \quad (30)$$

The former equation linearly relates the observed time series Y_t to the unobserved *state vector* α_t , and the latter gives the state evolution through a Markovian structure. The initial state vector is such that $\alpha_1 \sim N(\alpha_1, P_1)$. The remaining assumptions are that the random errors ε_t and η_t are independent (between each other and of α_1) and the *system matrices* Z_t, d_t, H_t, T_t, c_t and Q_t evolve deterministically. For a given time series of size T and any t, j , set $\mathcal{F}_j \equiv \sigma(Y_1, \dots, Y_j)$, $\hat{\alpha}_{t|j} \equiv E(\alpha_t | \mathcal{F}_j)$ and

$\hat{P}_{t|j} \equiv \text{Var}(\alpha_t | \mathcal{F}_j)$. The *Kalman filtering* consists of recursive equations for these first- and second-order conditional moments. The formulae and their respective deductions corresponding to *predicting* ($j = t - 1$), *filtering* ($j = t$) and *smoothing* ($j = T$), as details on the estimation of fixed unknown parameters (*hyperparameters*) in the system matrices by maximum likelihood, can be found in several books, from which we quote Brockwell and Davis [1], Harvey [8], Harvey [7], Hamilton [6], Durbin and Koopman [4], Brockwell and Davis [2] and Shumway and Stoffer [19].

The state space form for our multivariate SS stochastic drift model is presented in the sequel. The notation in equation (30) has been partially changed in order to match the corresponding adopted in the literature on oil price modeling (Schwartz and Smith [18]; Schwartz and Smith [17]).

- Measurement equation

$$\mathbf{Y}_t = \mathbf{B} + \mathbf{A}\Upsilon_t + v_t, \quad (31)$$

where

$$\mathbf{Y}_t = \begin{pmatrix} \ln F_{t, \tau_1}^{(1)} \\ \vdots \\ \ln F_{t, \tau_{h_1}}^{(1)} \\ \vdots \\ \ln F_{t, \tau_1}^{(1)} \\ \vdots \\ \ln F_{t, \tau_{h_d}}^{(d)} \\ \ln S_t^{(d+1)} \\ \vdots \\ \ln S_t^{(d+r)} \\ \ln F_{t, \tau_1}^{gnp} \\ \vdots \\ \ln F_{t, \tau_{h_{d+r+1}}}^{gnp} \end{pmatrix}, \mathbf{B} = \begin{pmatrix} B_1^*(\tau_1) \\ \vdots \\ B_1^*(\tau_{h_1}) \\ \vdots \\ B_d^*(\tau_1) \\ \vdots \\ B_d^*(\tau_{h_d}) \\ \phi_{d+1, 0} \\ \vdots \\ \phi_{d+r, 0} \\ B^{gnp*}(\tau_1) \\ \vdots \\ B^{gnp*}(\tau_{h_{d+r+1}}) \end{pmatrix}, v_t = \begin{pmatrix} v_{t, \tau_1}^{(1)} \\ \vdots \\ v_{t, \tau_{h_1}}^{(1)} \\ \vdots \\ v_{t, \tau_1}^{(d)} \\ \vdots \\ v_{t, \tau_{h_d}}^{(d)} \\ v_t^{(d+r)} \\ \vdots \\ v_{t, \tau_1}^{(d+r+1)} \\ \vdots \\ v_{t, \tau_{h_{d+r+1}}}^{(d+r+1)} \end{pmatrix}, \Upsilon_t = \begin{pmatrix} X_t \\ E_t \\ U_t \\ X_t^{gnp} \\ E_t^{gnp} \\ U_t^{gnp} \end{pmatrix}$$

$A =$

$$\begin{pmatrix} \phi_{1,1} e^{-\kappa_X \tau_1} & \phi_{1,2} & \frac{\phi_{1,2}}{\kappa_U} (1 - e^{-\kappa_U \tau_1}) & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \phi_{1,1} e^{-\kappa_X \tau_{h_1}} & \phi_{1,2} & \frac{\phi_{1,2}}{\kappa_U} (1 - e^{-\kappa_U \tau_{h_1}}) & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \phi_{d,1} e^{-\kappa_X \tau_1} & \phi_{d,2} & \frac{\phi_{d,2}}{\kappa_U} (1 - e^{-\kappa_U \tau_1}) & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \phi_{d,1} e^{-\kappa_X \tau_{h_d}} & \phi_{d,2} & \frac{\phi_{d,2}}{\kappa_U} (1 - e^{-\kappa_U \tau_{h_d}}) & 0 & 0 & 0 \\ \phi_{d+1,1} & \phi_{d+1,2} & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \phi_{d+r,1} & \phi_{d+r,2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & e^{\kappa_X^{gnp} \tau_1} & 1 & \frac{1}{\kappa_U^{gnp}} (1 - e^{-\kappa_U^{gnp} \tau_1}) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & e^{\kappa_X^{gnp} \tau_{h_d+r+1}} & 1 & \frac{1}{\kappa_U^{gnp}} (1 - e^{-\kappa_U^{gnp} \tau_{h_d+r+1}}) \end{pmatrix}$$

• State equation

$$\begin{pmatrix} X_t \\ E_t \\ U_t \\ X_t^{gnp} \\ E_t^{gnp} \\ U_t^{ng} \end{pmatrix} = \begin{pmatrix} e^{-\kappa_X \Delta t} & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & \Delta t & 0 & 0 & 0 \\ 0 & 0 & e^{-\kappa_U \Delta t} & 0 & 0 & 0 \\ 0 & 0 & 0 & e^{-\kappa_X^{gnp} \Delta t} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & \Delta t \\ 0 & 0 & 0 & 0 & 0 & e^{-\kappa_U^{gnp} \Delta t} \end{pmatrix} \begin{pmatrix} X_{t-\Delta t} \\ E_{t-\Delta t} \\ U_{t-\Delta t} \\ X_{t-\Delta t}^{gnp} \\ E_{t-\Delta t}^{gnp} \\ U_{t-\Delta t}^{ng} \end{pmatrix} + \begin{pmatrix} w_{1,t} \\ w_{2,t} \\ w_{3,t} \\ w_{4,t} \\ w_{5,t} \\ w_{6,t} \end{pmatrix}$$

$$\begin{pmatrix} w_{1,t} \\ w_{2,t} \\ w_{3,t} \\ w_{4,t} \\ w_{5,t} \\ w_{6,t} \end{pmatrix} \sim \mathbf{N} \left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} A & B \\ B' & D \end{pmatrix} \right), \quad (32)$$

where

$$A = \begin{pmatrix} \sigma_X^2 \Delta t & \rho_{XE} \sigma_X \sigma_E \Delta t & \rho_{XU} \sigma_X \sigma_U \Delta t \\ \rho_{XE} \sigma_X \sigma_E \Delta t & \sigma_E^2 \Delta t & \rho_{EU} \sigma_E \sigma_U \Delta t \\ \rho_{XU} \sigma_X \sigma_U \Delta t & \rho_{EU} \sigma_E \sigma_U \Delta t & \sigma_U^2 \Delta t \end{pmatrix}$$

$$B = \begin{pmatrix} \rho_{XX} gnp \sigma_X \sigma_X gnp \Delta t & \rho_{XE} gnp \sigma_X \sigma_E gnp \Delta t & \rho_{XU} gnp \sigma_X \sigma_U gnp \Delta t \\ \rho_{EX} gnp \sigma_E \sigma_X gnp \Delta t & \rho_{EE} gnp \sigma_E \sigma_E gnp \Delta t & \rho_{EU} gnp \sigma_E \sigma_U gnp \Delta t \\ \rho_{UX} gnp \sigma_U \sigma_X gnp \Delta t & \rho_{UE} gnp \sigma_U \sigma_E gnp \Delta t & \rho_{UU} gnp \sigma_U \sigma_U gnp \Delta t \end{pmatrix}$$

$$C = \begin{pmatrix} (\sigma_X gnp)^2 \Delta t & \rho_X gnp E gnp \sigma_X gnp \sigma_E gnp \Delta t \\ \rho_X gnp X gnp \sigma_E gnp \Delta t & (\sigma_X gnp)^2 \Delta t \\ \rho_X gnp U gnp \sigma_X gnp \sigma_U gnp \Delta t & \rho_E gnp U gnp \sigma_E gnp \sigma_U gnp \Delta t \\ \rho_X gnp U gnp \sigma_X gnp \Delta t & \rho_{EU} gnp \sigma_E gnp \sigma_U gnp \Delta t \\ (\sigma_U gnp)^2 \Delta t \end{pmatrix}$$

5. Results

5.1. Estimation

The SS multivariate stochastic drift model will be estimated using monthly time series of prices of oil and several refined products (in US\$)in the period from January 1985 to September 2009. The products considered are detailed in Table 2. Futures prices for crude oil (WTI), gasoline and natural gas have been obtained from NYMEX (<http://www.cmegroup.com>), whereas for the remaining products, with no future markets, Platts (<http://www.platts.com>) has been the source. In order to eliminate the potential contribution of the US inflation to the long run trend specified in the model, it was decided to previously convert nominal prices to real prices using the Producer Price Index (PPI) as given by the Bureau of Labor Statistics (BLS, <http://data.bls.gov/cgi-bin/dsrv?wp>).

Table 2. Oil and refined products considered in the multivariate model.

Product	Price type	Unit of measurement
oil (WTI)	1, 2, 4, 3, 6 months futures	USD/bbl
gasoline	1, 2, 4, 3, 6 months futures	USD/bbl
jet fuel	Platts	USD/bbl
heating oil (LSFO)	Platts	USD/bbl
naphtha	Platts	USD/bbl
diesel	Platts	USD/bbl
propane	Platts	USD/bbl
butane	Platts	USD/bbl
propene	Platts	USD/bbl
natural gas	1, 24, 36 months futures	USD/MMBtu

Our multivariate model has a huge number of hiperparameters to be estimated (at most 79 hiperparameters). After some experimentation with likelihood full maximization, which was taking too much time to converge, we have decided to split the optimization process in three stages, in order to improve computational efficiency. In the first stage it was estimated just the hiperparameters associated with the natural gas factors, e.g., volatilities, mean reversion parameter etc. Then, given the parameters estimated in the first stage, we proceeded to the second stage, where hiperparameters from the WTI factors and from the affine functions linking the product factors to the global factors were estimated. Finally, given the estimates obtained in the first and second stages, we obtained the estimates of the covariances between the shocks that drive WTI and natural gas factors (6 factors all together). In Table 3 we show the parameters estimates obtained.

5.1.1. Simulation

A natural motivation for derivations and estimations of this multivariate model is to develop and implement a simulator for oil and refined products spot prices, within the scope of a corporate ALM project for the oil industry. In particular, the scope of the project requires prices simulations for very long

horizons (up to 50 years). From the outset, it is clear that models like ours, with stochastic trends, may produce simulated long term prices with values too far away from what could be considered acceptable by oil market analysts. Figure 3 depicts spot prices simulated for oil (WTI) for the next 30 years, obtained from our multivariate model.

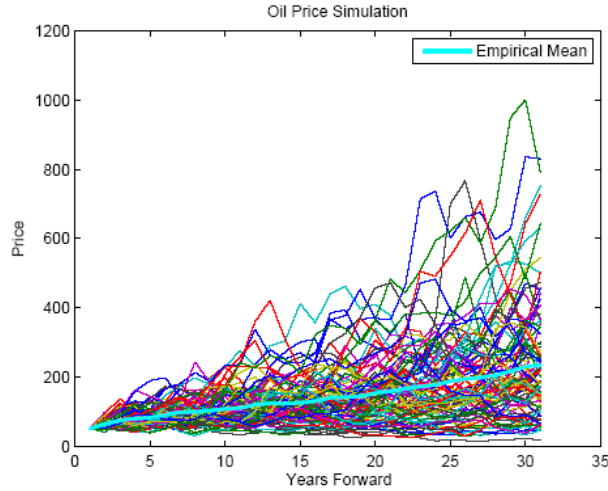


Figure 3. WTI simulated spot prices 30 years ahead (with no calibration).

A common attitude in the oil market is to believe in the existence of a “long run equilibrium price” for oil, this being the result of the interplay between the forces of demand and supply that operate in such markets. Therefore, in order to simulate prices coherent with such beliefs, it would be necessary to previously calibrate some of the fixed parameters and/or final state estimates obtained by the fitting of our multivariate model. A clue for the operationalization of this procedure is given by the form of the forecasting function for WTI spot price, as given in equation 28 (consider $\phi_{1,0} = 0$, $\phi_{1,1} = 1$ and $\phi_{1,2} = 1$). As it can be seen this function is essentially the exponential of a deterministic linear trend. As such, if one wants to attenuate the trend, and therefore, obtain simulated prices coherent with an “equilibrium price”, calibration should be undertaken on the parameters associated with the slope, namely $\hat{\sigma}_X$, $\hat{\sigma}_E$, $\hat{\sigma}_U$ and $\bar{U}$ (note that this same strategy may also be adopted to calibrate simulations for natural gas spot prices).

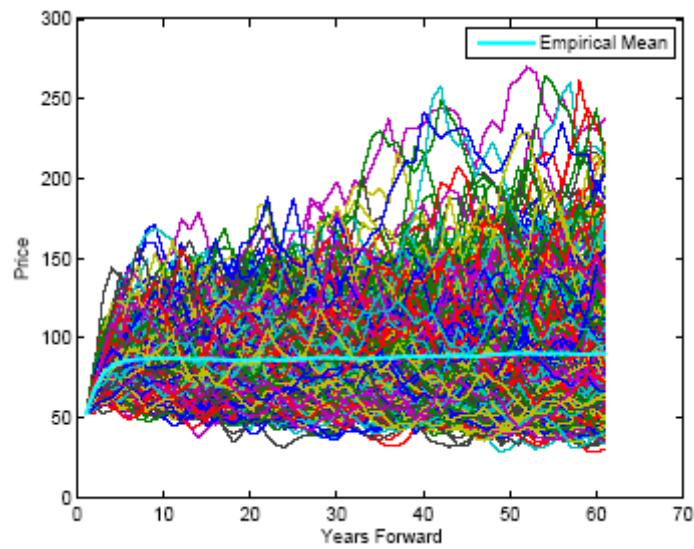
Table 3. Parameters estimates for the multivariate model for crude oil and refined products.

Parameter	estimate	parameter	estimate	parameter	estimate	parameter	estimate
σ_X	0.124	$(\bar{U}^{gnp*} - \lambda_E^{gnp})$	-1.148	$\phi_{3,1}$	1.425	$\phi_{10,0}$	0.000
σ_E	0.307	$\rho_{X^{gnp}E^{gnp}}$	-0.444	$\phi_{3,2}$	1.000	$\phi_{10,1}$	1.000
σ_U	0.214	$\rho_{X^{gnp}U^{gnp}}$	0.426	$\phi_{4,0}$	0.000	$\phi_{10,2}$	1.000
κ_X	0.238	$\rho_{E^{gnp}U^{gnp}}$	-0.900	$\phi_{4,1}$	1.000	$\sigma_{v_{\tau_1},1}$	0.056
κ_U	0.696	$\rho_{X^{gnp}X}$	-0.192	$\phi_{4,2}$	0.962	$\sigma_{v_{\tau_2},1}$	0.000
λ_X	12.079	$\rho_{X^{gnp}E}$	-0.050	$\phi_{5,0}$	0.189	$\sigma_{v_{\tau_3},1}$	0.032
$\bar{U}$	0.029	$\rho_{X^{gnp}U}$	-0.025	$\phi_{5,1}$	1.193	$\sigma_{v_{\tau_1},2}$	0.101
$\bar{U}^*$	-8.231	$\rho_{E^{gnp}X}$	0.234	$\phi_{5,2}$	1.000	$\sigma_{v_{\tau_2},2}$	0.074
$(\bar{U}^* - \lambda_E)$	5.159	$\rho_{E^{gnp}E}$	0.277	$\phi_{6,0}$	0.212	$\sigma_{v_{\tau_3},2}$	0.061
ρ_{XE}	-0.424	$\rho_{E^{gnp}U}$	-0.268	$\phi_{6,1}$	1.367	$\sigma_{v,3}$	0.025
ρ_{XU}	0.344	$\rho_{U^{gnp}X}$	-0.144	$\phi_{6,2}$	1.000	$\sigma_{v,4}$	0.126
ρ_{EU}	-0.883	$\rho_{U^{gnp}E}$	-0.336	$\phi_{7,0}$	-0.301	$\sigma_{v,5}$	0.088
σ_X^{gnp}	0.507	$\rho_{U^{gnp}U}$	0.375	$\phi_{7,1}$	-0.054	$\sigma_{v,6}$	0.017
σ_E^{gnp}	0.380	$\phi_{1,0}$	0.000	$\phi_{7,2}$	1.000	$\sigma_{v,7}$	0.071
σ_U^{gnp}	0.216	$\phi_{1,1}$	1.000	$\phi_{8,0}$	-0.099	$\sigma_{v,8}$	0.042
κ_X^{gnp}	1.329	$\phi_{1,2}$	1.000	$\phi_{8,1}$	0.170	$\sigma_{v,9}$	0.197
κ_U^{gnp}	0.296	$\phi_{2,0}$	0.199	$\phi_{8,2}$	1.000	$\sigma_{v_{\tau_1},10}$	0.000
λ_X^{gnp}	1.881	$\phi_{2,1}$	1.005	$\phi_{9,0}$	0.625	$\sigma_{v_{\tau_2},10}$	0.000
$\bar{U}^{gnp}$	0.060	$\phi_{2,2}$	1.000	$\phi_{9,1}$	1.000	$\sigma_{v_{\tau_3},10}$	0.000
$\bar{U}^{gnp*}$	-2.175	$\phi_{3,0}$	0.261	$\phi_{9,2}$	1.000		

Table 4. Descriptive statistics for WTI simulated spot prices (with no calibration).

US\$/bbl (Sept 2009)	10 years (2021)	20 years (2031)	30 years (2041)
Mean	135, 2	207, 8	309, 1
Median	117, 1	164, 8	218, 6
Q-75%	164, 9	257, 4	378, 0
Q-95%	396, 4	794, 4	1.505, 0

After this calibration was performed, we simulated some long run prices for WTI and some of its derived products. The results are shown in the sequel.



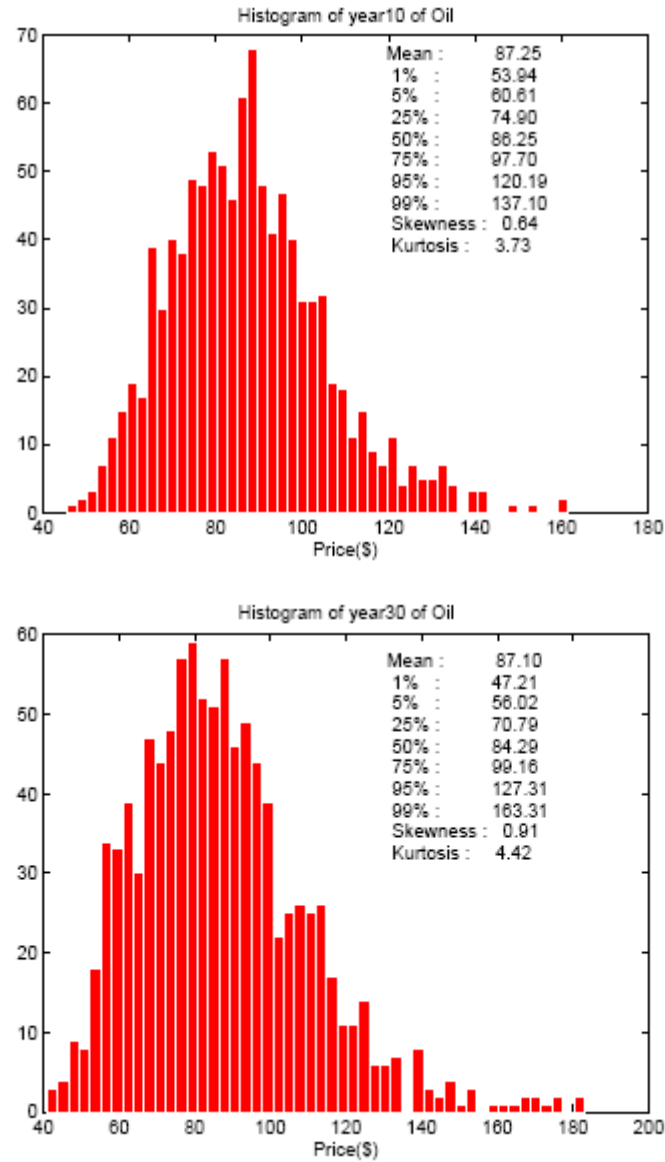
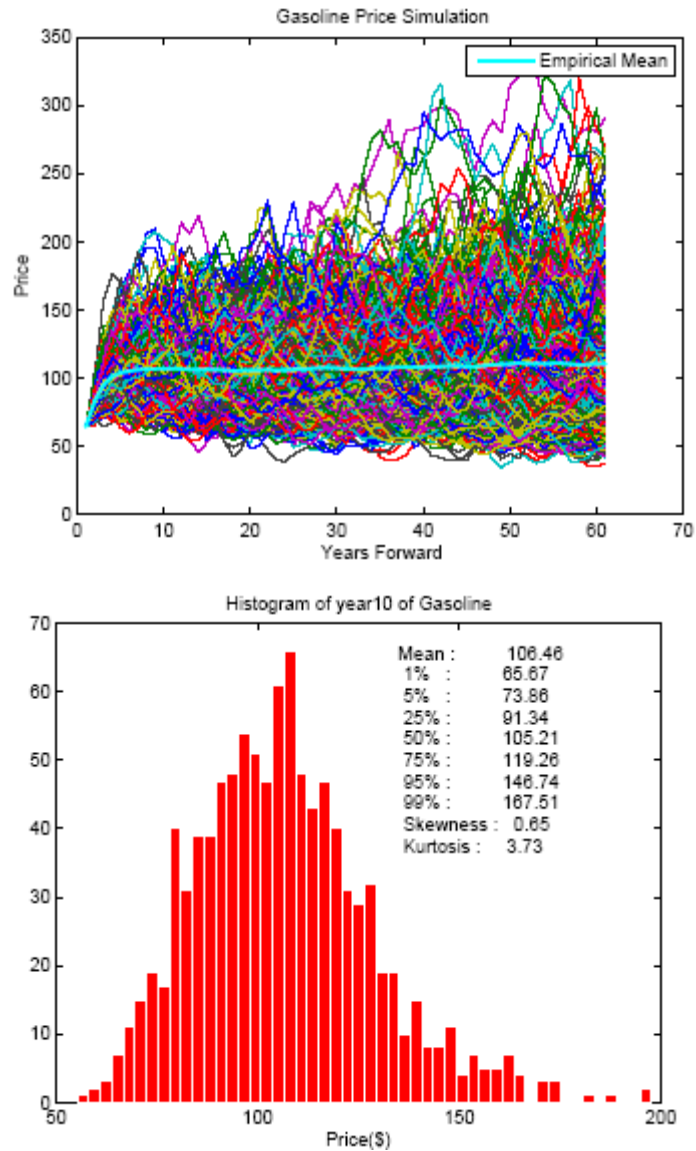


Figure 4. WTI simulated spot prices and descriptive statistics 10 and 30 years ahead based on equilibrium price at US\$ 88, 00/bbl (median).



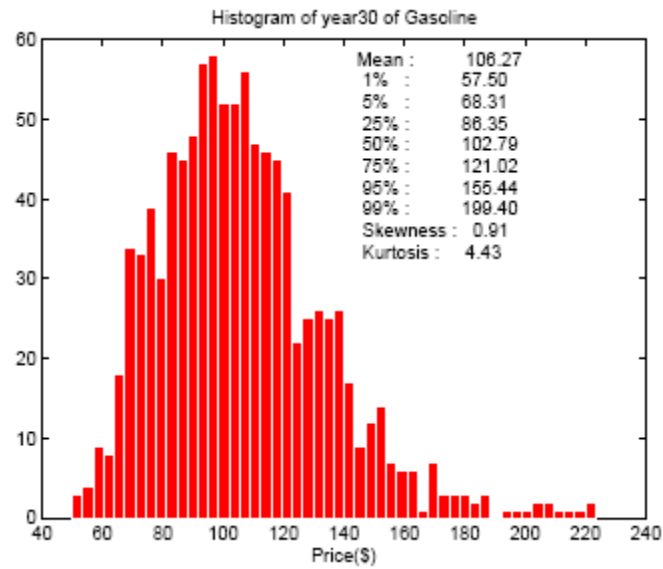
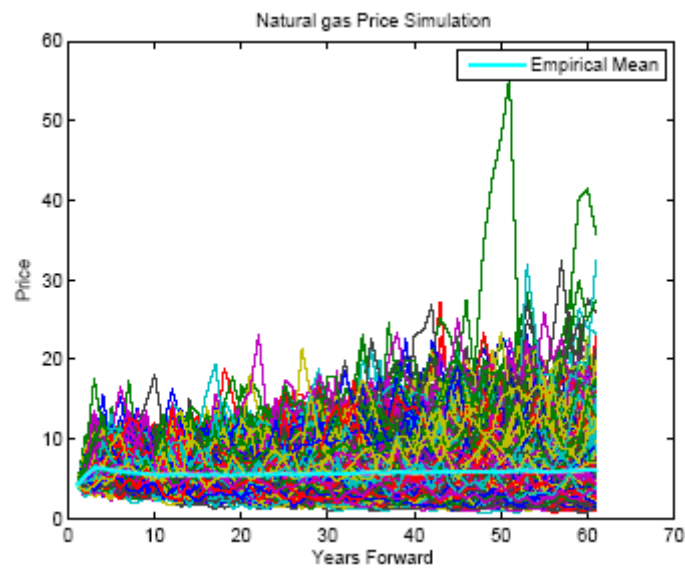


Figure 5. Gasoline simulated spot prices and descriptive statistics 10 and 30 years ahead based on WTI equilibrium price at US\$ 88, 00/bbl (median).



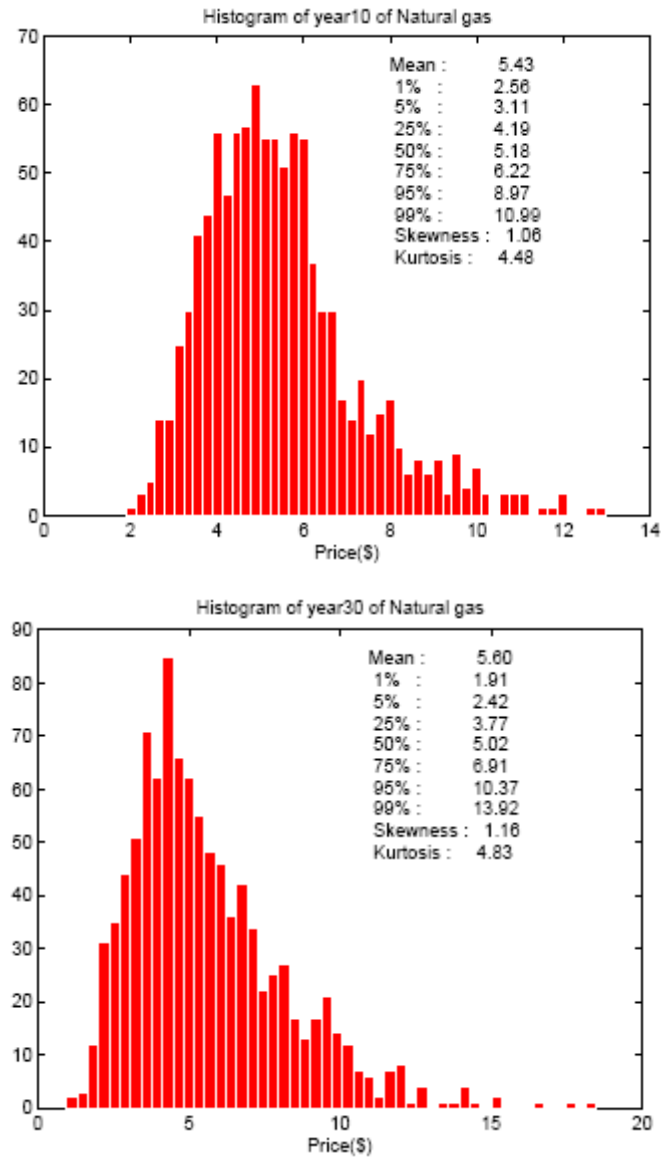
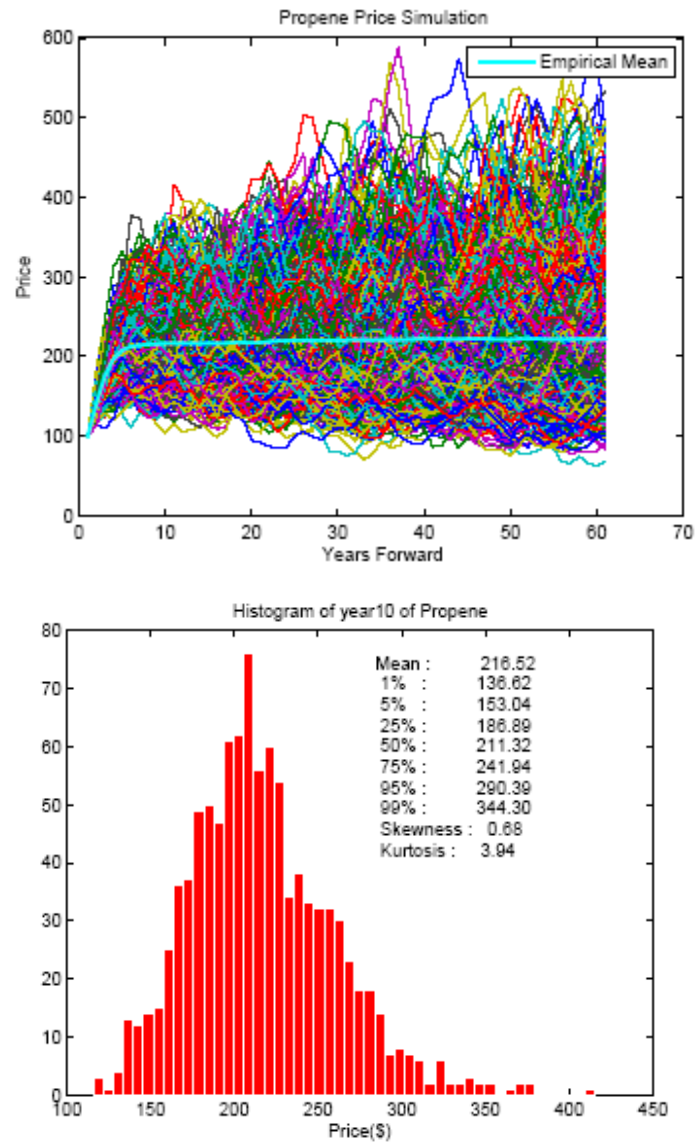


Figure 6. Natural gas simulated spot prices and descriptive statistics 10 and 30 years ahead based on an equilibrium price at US\$ 5,00/mmBTU (median).



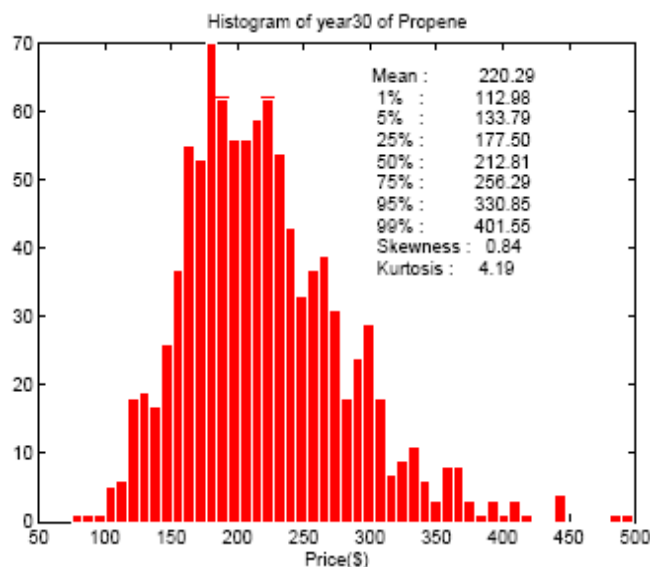


Figure 7. Propene simulated spot prices and descriptive statistics 10 and 30 years ahead based on WTI equilibrium price at US\$ 88, 00/bbl (median).

6. Conclusions

We have developed and estimated a multivariate state space model for the joint forecasting and simulation of oil and refined products prices. Our model is based on Schwartz and Smith [18] two-factor model and the multivariate framework used in Diebold et al. [5]. The resulting model has two pairs of factors, each pair representing long term and short term components that drive the prices of oil and nine of its derived products. Such parsimony can be justified on two grounds: empirical support given in the literature and operational feasibility.

Our approach has proved able to forecasting and simulating both spot and future prices at different maturities. More specifically the model is able to generate spot prices for these products, and thanks to calibration, simulated values can be obtained from a flexible forecasting function, which can express a long run trend and an equilibrium value for the long run prices. This is in tune with the shapes of the main trend models for prices in the oil industry, namely, the random walk and the mean reversion. Refinements in this direction can be accomplished by extending Nyblom and Harvey [15] test for the number of common stochastic trends in a vector of time series.

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